The Theory of Irrational Return and Risk

An Interdisciplinary Framework in Algebra, Financial Economics, Differential Geometry, Theoretical Econometrics, Dynamic Systems, Statistics and Computational Science

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Abstract

This treatise develops a mathematical and interdisciplinary framework for the structural relationship between Irrational Return, denoted by r, and Risk, denoted by R.

The fundamental relationship is specified through two exponential-reciprocal functions:

$$ae^{r} + b + ce^{-r} = fe^{R} + g + he^{-R}. \tag{1}$$

The framework investigates the algebraic solutions, existence conditions, multiplicity of equilibrium branches, structural balance points, differential properties, Jacobian, Hessian, curvature, singularities, local invertibility and comparative statics.

The analysis is subsequently extended to financial economics, portfolio theory, wealth accumulation, volatility measurement, equilibrium theory, nonlinear econometrics, parameter identification, generalized method of moments, maximum likelihood, panel data, structural breaks and stochastic volatility.

Further extensions investigate continuous-time dynamics, stochastic differential equations, control theory, computational algorithms, artificial intelligence, cybersecurity, institutional economics, law, political economy, international relations, welfare economics, geopolitical risk and warfare economics.

The distinction between mathematical propositions, economic interpretations, empirical hypotheses and conjectures is maintained throughout.

The principal objective is to establish a general structural framework through which nonlinear relationships between financial return and risk can be represented, analysed, estimated and tested.

The treatise ends with "The End"

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1 Introduction

1.1 Motivation

Financial economics has traditionally investigated relationships between return, risk, asset prices, portfolio allocation and intertemporal wealth.

The foundational contributions of Markowitz, Sharpe, Lintner and Merton established important mathematical relationships between expected return, variance, systematic risk and portfolio choice [2, 3, 4, 5].

However, the relationship between return and risk need not always be represented adequately by a globally linear function.

Financial markets exhibit nonlinearities, asymmetric responses, volatility clustering, structural breaks, regime changes and potentially multiple equilibrium configurations.

The present theory proposes an exponential-reciprocal representation.

$$\boxed{ae^{r} + b + ce^{-r} = fe^{R} + g + he^{-R}} \tag{2}$$

The variables are:

r Irrational Return.

R Risk.

a, c Exponential and reciprocal-return coefficients.

f, h Exponential and reciprocal-risk coefficients.

b, g Additive structural coefficients.

The adjective Irrational is used as the name of the theoretical construct. It does not imply that r must be an irrational number in the mathematical number-theoretic sense.

1.2 Research objectives

The principal objectives are:

1. Establish the algebraic foundations.
2. Derive return as a function of risk.
3. Derive risk as a function of return.
4. Establish existence and multiplicity conditions.
5. Investigate structural balance points.
6. Derive first and second derivatives.
7. Investigate geometric properties of equilibrium.
8. Develop financial-economic interpretations.
9. Establish econometric identification conditions.
10. Construct statistical estimation procedures.
11. Develop dynamic and stochastic extensions.
12. Identify applications across other disciplines.

2 Algebraic Foundations

2.1 Structural functions

Define the Irrational Return Function:

$$\mathcal{I}(r) = ae^{r} + b + ce^{-r}. \tag{3}$$

Define the Risk Function:

$$\mathcal{K}(R) = fe^{R} + g + he^{-R}. \tag{4}$$

The equilibrium condition becomes:

$$\mathcal{I}(r) = \mathcal{K}(R). \tag{5}$$

We initially assume:

$$a, b, c, f, g, h \in \mathbb{R}. \tag{6}$$

$$r, R \in \mathbb{R}. \tag{7}$$

2.2 Transformation into quadratic equations

Introduce:

$$x = e^{r}, \qquad y = e^{R}. \tag{8}$$

Since exponential functions are positive:

$$x > 0, \qquad y > 0. \tag{9}$$

The original equation becomes:

$$ax + b + \frac{c}{x} = fy + g + \frac{h}{y}. \tag{10}$$

Rearranging and multiplying by x:

$$ax^2 - \left(fy + g + \frac{h}{y} - b\right)x + c = 0. \tag{11}$$

This is quadratic in x.
Similarly, multiplying by y:

$$fy^2 - \left(ax + b + \frac{c}{x} - g\right)y + h = 0. \tag{12}$$

This is quadratic in y.

Thus, the exponential-reciprocal model is algebraically reducible to two quadratic equations in positive variables.

2.3 Solution for Irrational Return

Define:

$$\Phi(R) = fe^R + g + he^{-R} - b. \tag{13}$$

Then:

$$ae^{2r} - \Phi(R)e^r + c = 0. \tag{14}$$

Applying the quadratic formula:

$$e^r = \frac{\Phi(R) \pm \sqrt{\Phi(R)^2 - 4ac}}{2a}. \tag{15}$$

Therefore:

$$\boxed{r_\pm(R) = \ln\left[\frac{\Phi(R) \pm \sqrt{\Phi(R)^2 - 4ac}}{2a}\right].} \tag{16}$$

The discriminant is:

$$\Delta_r(R) = \Phi(R)^2 - 4ac. \tag{17}$$

For real solutions:

$$\Delta_r(R) \geq 0. \tag{18}$$

Additionally, the argument of the logarithm must be strictly positive.

2.4 Solution for Risk

Define:

$$\Psi(r) = ae^r + b + ce^{-r} - g. \tag{19}$$

Then:

$$fe^{2R} - \Psi(r)e^R + h = 0. \tag{20}$$

Applying the quadratic formula:

$$e^R = \frac{\Psi(r) \pm \sqrt{\Psi(r)^2 - 4fh}}{2f}. \tag{21}$$

Therefore:

$$\boxed{R_\pm(r) = \ln\left[\frac{\Psi(r) \pm \sqrt{\Psi(r)^2 - 4fh}}{2f}\right].} \tag{22}$$

The discriminant is:

$$\Delta_R(r) = \Psi(r)^2 - 4fh. \tag{23}$$

The conditions are:

$$\Delta_R(r) \geq 0 \tag{24}$$

and positivity of the logarithm argument.

2.5 Existence and multiplicity

Proposition 2.1. *For nonzero a, a given R admits at most two real-valued solutions for r.*

For nonzero f, a given r admits at most two real-valued solutions for R.

Proof. The equations for e^r and e^R are quadratic.

A quadratic polynomial has at most two real roots.

Since the exponential transformations are bijections from the real line to the positive real numbers, each admissible positive root corresponds to exactly one real logarithmic solution.

Therefore, each conditional solution set contains at most two elements. □

2.6 Degenerate cases

If $a = 0$, the equation for e^r becomes linear provided the remaining coefficient is nonzero.

If $c = 0$, the quadratic has a zero root that is inadmissible because $e^r > 0$.

Analogous considerations apply when $f = 0$ or $h = 0$.

These cases must be treated separately rather than blindly applying the quadratic formula.

3 Structural Balance and Hyperbolic Geometry

Assume:

$$a, c, f, h > 0. \tag{25}$$

3.1 Return balance point

Differentiate the return function:

$$\mathcal{I}'(r) = ae^r - ce^{-r}. \tag{26}$$

The stationary condition is:

$$ae^r = ce^{-r}. \tag{27}$$

Therefore:

$$\boxed{r^* = \frac{1}{2}\ln\left(\frac{c}{a}\right).} \tag{28}$$

The second derivative is:

$$\mathcal{I}''(r) = ae^r + ce^{-r} > 0. \tag{29}$$

Thus, r^* is the unique global minimum.
The minimum value is:

$$\boxed{\mathcal{I}_{\min} = b + 2\sqrt{ac}.} \tag{30}$$

3.2 Risk balance point

Similarly:

$$\mathcal{K}'(R) = fe^{R} - he^{-R}. \tag{31}$$

The stationary condition gives:

$$\boxed{R^* = \frac{1}{2}\ln\left(\frac{h}{f}\right).} \tag{32}$$

Since:

$$\mathcal{K}''(R) = fe^{R} + he^{-R} > 0, \tag{33}$$

the risk function is strictly convex.
Its minimum is:

$$\boxed{\mathcal{K}_{\min} = g + 2\sqrt{fh}.} \tag{34}$$

3.3 Hyperbolic representation

Recall:

$$\cosh z = \frac{e^{z} + e^{-z}}{2}. \tag{35}$$

Using the balance point:

$$ae^{r} + ce^{-r} = 2\sqrt{ac}\cosh(r - r^*). \tag{36}$$

Therefore:

$$\boxed{\mathcal{I}(r) = b + 2\sqrt{ac}\cosh(r - r^*).} \tag{37}$$

Similarly:

$$\boxed{\mathcal{K}(R) = g + 2\sqrt{fh}\cosh(R - R^*).} \tag{38}$$

The fundamental equation becomes:

$$\boxed{b + 2\sqrt{ac}\cosh(r - r^*) = g + 2\sqrt{fh}\cosh(R - R^*).} \tag{39}$$

This is the hyperbolic representation of the Irrational Return–Risk relationship.

Theorem 3.1. *If $a, c, f, h > 0$, both structural functions are strictly convex and possess unique global minima.*

Proof. Their second derivatives are strictly positive everywhere.

Furthermore, both functions tend to positive infinity as their respective arguments tend to either positive or negative infinity.

Consequently, each function possesses exactly one global minimum. □

3.4 Reflection symmetry

If $a = c$, then:

$$r^* = 0. \tag{40}$$

The return function becomes:

$$\mathcal{I}(r) = 2a\cosh(r) + b. \tag{41}$$

It satisfies:

$$\mathcal{I}(r) = \mathcal{I}(-r). \tag{42}$$

Likewise, if $f = h$:

$$\mathcal{K}(R) = \mathcal{K}(-R). \tag{43}$$

These conditions establish reflection symmetry around zero.

3.5 Existence of equilibrium

Define:

$$A = b + 2\sqrt{ac}. \tag{44}$$

$$B = g + 2\sqrt{fh}. \tag{45}$$

Under positive coefficients:

$$\text{Range}(\mathcal{I}) = [A, \infty). \tag{46}$$

$$\text{Range}(\mathcal{K}) = [B, \infty). \tag{47}$$

Theorem 3.2. *A real equilibrium pair always exists under the positive-coefficient assumptions.*

Proof. Both ranges are unbounded above and have finite lower bounds.

Their intersection is:

$$[A, \infty) \cap [B, \infty) = [\max(A, B), \infty). \tag{48}$$

This intersection is nonempty.

Therefore, at least one pair (r, R) satisfies the equilibrium equation. □

A stronger result concerns singular equilibrium.

Theorem 3.3. *The equilibrium manifold contains a singular point if and only if:*

$$\boxed{b + 2\sqrt{ac} = g + 2\sqrt{fh}.} \tag{49}$$

Proof. A singular point requires both first derivatives to vanish.

This uniquely determines r^* and R^*.

Substituting these values into the equilibrium equation gives:

$$b + 2\sqrt{ac} = g + 2\sqrt{fh}. \tag{50}$$

The condition is therefore both necessary and sufficient. □

4 Differential Theory

4.1 First derivatives

Define:

$$F(r, R) = \mathcal{I}(r) - \mathcal{K}(R). \tag{51}$$

Then:

$$F(r, R) = 0. \tag{52}$$

The first partial derivatives are:

$$F_r = ae^r - ce^{-r}, \tag{53}$$
$$F_R = -fe^R + he^{-R}. \tag{54}$$

Implicit differentiation gives:

$$F_r\, dr + F_R\, dR = 0. \tag{55}$$

Hence:

$$\boxed{\frac{dR}{dr} = \frac{ae^r - ce^{-r}}{fe^R - he^{-R}}.} \tag{56}$$

Conversely:

$$\boxed{\frac{dr}{dR} = \frac{fe^R - he^{-R}}{ae^r - ce^{-r}}.} \tag{57}$$

These expressions apply where their denominators are nonzero.

4.2 Second derivatives

Differentiating the equilibrium condition twice:

$$\mathcal{I}''(r) = \mathcal{K}''(R)\left(\frac{dR}{dr}\right)^2 + \mathcal{K}'(R)\frac{d^2R}{dr^2}. \tag{58}$$

Therefore:

$$\boxed{\frac{d^2R}{dr^2} = \frac{\mathcal{I}''(r) - \mathcal{K}''(R)\left(\frac{dR}{dr}\right)^2}{\mathcal{K}'(R)}.} \tag{59}$$

Substituting:

$$\boxed{\frac{d^2R}{dr^2} = \frac{ae^r + ce^{-r} - (fe^R + he^{-R})\left(\frac{dR}{dr}\right)^2}{fe^R - he^{-R}}.} \tag{60}$$

The curvature depends on the relative marginal and second-order structural effects.

4.3 Jacobian

The Jacobian is:

$$J(r,R) = \begin{pmatrix} ae^r - ce^{-r} & -fe^R + he^{-R} \end{pmatrix}. \tag{61}$$

A regular point satisfies:

$$J(r,R) \neq (0,0). \tag{62}$$

At regular points, the equilibrium set is locally a smooth one-dimensional manifold. At singular points, the ordinary implicit function theorem does not apply.

4.4 Hessian

The Hessian is:

$$H_F(r,R) = \begin{pmatrix} ae^r + ce^{-r} & 0 \\ 0 & -fe^R - he^{-R} \end{pmatrix}. \tag{63}$$

Its determinant is:

$$\det(H_F) = -(ae^r + ce^{-r})(fe^R + he^{-R}). \tag{64}$$

Under positive coefficients:

$$\det(H_F) < 0. \tag{65}$$

The Hessian is indefinite.

Thus, the structural function has convexity in the return coordinate and concavity in the risk coordinate.

5 Equilibrium Manifold and Branch Geometry

5.1 Regular branches

At a point where:

$$F_r \neq 0, \tag{66}$$

return can locally be expressed as a function of risk.
At a point where:

$$F_R \neq 0, \tag{67}$$

risk can locally be expressed as a function of return.
These are distinct local invertibility conditions.

5.2 Branch configurations

The relative values of A and B determine the global configuration.

If $A > B$, the risk function must attain values above its minimum to match every value of the return function. For every r, there are two risk branches.

If $A < B$, the corresponding statement holds with return and risk interchanged.

If $A = B$, the two functions attain their minima at the same equilibrium value.

Proposition 5.1. *Under positive coefficients:*

1. *If $A > B$, the equilibrium set has two disconnected branches that can be represented as $R_+(r)$ and $R_-(r)$.*
2. *If $A < B$, it has two disconnected branches that can be represented as $r_+(R)$ and $r_-(R)$.*
3. *If $A = B$, the two branches intersect at the singular balance point.*

5.3 Local bifurcation

When $A = B$, Taylor expansion around the balance points gives:

$$\begin{aligned}\mathcal{I}(r) = A &+ \sqrt{ac}(r - r^*)^2 \\ &+ \frac{\sqrt{ac}}{12}(r - r^*)^4 + O((r - r^*)^6).\end{aligned} \tag{68}$$

Similarly:

$$\begin{aligned}\mathcal{K}(R) = A &+ \sqrt{fh}(R - R^*)^2 \\ &+ \frac{\sqrt{fh}}{12}(R - R^*)^4 + O((R - R^*)^6).\end{aligned} \tag{69}$$

The leading-order equilibrium condition is:

$$\sqrt{ac}(r - r^*)^2 = \sqrt{fh}(R - R^*)^2. \tag{70}$$

Therefore:

$$\boxed{R - R^* \approx \pm \left(\frac{ac}{fh}\right)^{1/4} (r - r^*).} \tag{71}$$

The two branches intersect with distinct tangent directions.

This is a local crossing of equilibrium branches.

6 Financial Economics

6.1 Logarithmic return

Let P_0 and P_1 denote initial and terminal asset prices.

Define continuously compounded return:

$$r = \ln\left(\frac{P_1}{P_0}\right). \tag{72}$$

Then:

$$e^r = \frac{P_1}{P_0}. \tag{73}$$

The Irrational Return Function becomes:

$$\boxed{\mathcal{I}(r) = a\frac{P_1}{P_0} + b + c\frac{P_0}{P_1}.} \tag{74}$$

The function combines gross appreciation, a structural baseline and reciprocal gross price movement.

The reciprocal term permits an asymmetric representation of appreciation and contraction.

6.2 Wealth interpretation

Let W_0 and W_1 represent initial and terminal wealth.

Then:

$$r = \ln\left(\frac{W_1}{W_0}\right). \tag{75}$$

The structural function becomes:

$$\mathcal{I}(r) = a\frac{W_1}{W_0} + b + c\frac{W_0}{W_1}. \tag{76}$$

This provides a nonlinear representation of wealth accumulation and wealth contraction.

It is not itself a utility function or a measure of investor welfare.

Those interpretations require additional assumptions.

6.3 Volatility representation of risk

Let $\sigma > 0$ denote volatility and let $\sigma_0 > 0$ be a reference volatility.

Define:

$$R = \ln\left(\frac{\sigma}{\sigma_0}\right). \tag{77}$$

Therefore:

$$e^R = \frac{\sigma}{\sigma_0}. \tag{78}$$

The Risk Function becomes:

$$\boxed{\mathcal{K}(R) = f\frac{\sigma}{\sigma_0} + g + h\frac{\sigma_0}{\sigma}.} \tag{79}$$

The resulting financial relationship is:

$$\boxed{a\frac{P_1}{P_0} + b + c\frac{P_0}{P_1} = f\frac{\sigma}{\sigma_0} + g + h\frac{\sigma_0}{\sigma}.} \tag{80}$$

This is an operational empirical specification.

Volatility is only one possible risk proxy. Alternative measures include:

Variance Second central moment of returns.

Downside deviation Dispersion below a specified threshold.

Expected shortfall Conditional expected loss beyond a specified quantile.

Credit risk Exposure to default and recovery risk.

Liquidity risk Exposure to inability to transact without significant price impact.

Systemic risk Exposure to instability across interconnected financial institutions.

The chosen risk measure must be explicitly identified in every empirical application.

6.4 Portfolio extension

For n assets, define portfolio log return under a specified portfolio-weighting convention:

$$r_p = \sum_{i=1}^{n} w_i r_i. \tag{81}$$

For a fully invested portfolio:

$$\sum_{i=1}^{n} w_i = 1. \tag{82}$$

Then:

$$\mathcal{I}(r_p) = ae^{\sum_{i=1}^{n} w_i r_i} + b + ce^{-\sum_{i=1}^{n} w_i r_i}. \tag{83}$$

This is a nonlinear transformation of aggregate portfolio return.

For a buy-and-hold portfolio with initial weights w_i, the exact terminal portfolio gross return is:

$$G_p = \sum_{i=1}^{n} w_i e^{r_i}. \tag{84}$$

Consequently, the exact portfolio log return is:

$$r_p = \ln\left(\sum_{i=1}^{n} w_i e^{r_i}\right). \tag{85}$$

This distinction is important when implementing the theory with actual portfolio data.

6.5 Financial planning and analysis

The model can be used as a structural component of financial planning.

Potential applications include:

1. Wealth accumulation scenarios.
2. Portfolio stress testing.
3. Capital allocation.
4. Risk-budget determination.
5. Sensitivity analysis.
6. Scenario-based financial forecasting.
7. Solvency and liquidity planning.

For example, a financial planner may investigate the set of admissible returns associated with a predetermined risk constraint.

However, the theory alone does not determine the optimal portfolio.

A complete optimization problem would require an objective function, investment constraints, cash-flow assumptions and an appropriate risk preference specification.

7 Comparative Statics

Recall:

$$F(r, R) = 0. \tag{86}$$

At regular points, differentiating with respect to a structural parameter θ, while holding R fixed, gives:

$$\frac{\partial r}{\partial \theta} = -\frac{F_\theta}{F_r}. \tag{87}$$

The parameter derivatives are:

$$F_a = e^r, \tag{88}$$
$$F_b = 1, \tag{89}$$
$$F_c = e^{-r}, \tag{90}$$
$$F_f = -e^R, \tag{91}$$
$$F_g = -1, \tag{92}$$
$$F_h = -e^{-R}. \tag{93}$$

Therefore:

Table 1: Comparative statics of return

Parameter	Derivative	Interpretation
a	$-e^r/F_r$	Return coefficient effect
b	$-1/F_r$	Additive return effect
c	$-e^{-r}/F_r$	Reciprocal return effect
f	e^R/F_r	Exponential risk effect
g	$1/F_r$	Additive risk effect
h	e^{-R}/F_r	Reciprocal risk effect

For risk, holding r fixed:

$$\frac{\partial R}{\partial \theta} = -\frac{F_\theta}{F_R}. \tag{94}$$

These expressions provide local comparative statics for structural changes.

8 Elasticity and Financial Sensitivity

The marginal risk-return relationship is:

$$\frac{dR}{dr} = \frac{ae^r - ce^{-r}}{fe^R - he^{-R}}. \tag{95}$$

If R is logarithmic volatility:

$$R = \ln(\sigma/\sigma_0), \tag{96}$$

then:

$$\frac{dR}{dr} = \frac{d\ln\sigma}{dr}. \tag{97}$$

Consequently:

$$\boxed{\frac{d\ln\sigma}{dr} = \frac{ae^r - ce^{-r}}{fe^R - he^{-R}}.} \tag{98}$$

This is the local percentage sensitivity of volatility to a change in continuously compounded return along the equilibrium manifold.

The conventional elasticity:

$$\mathcal{E}_{R,r} = \frac{r}{R}\frac{dR}{dr} \tag{99}$$

is meaningful only where r and R are nonzero and the ratio-scale interpretation is justified.

9 Theoretical Econometrics

9.1 Stochastic structural specification

The deterministic equation is unlikely to hold exactly for observed financial data.

Introduce a disturbance:

$$ae^{r_t} + b + ce^{-r_t} = fe^{R_t} + g + he^{-R_t} + \varepsilon_t. \tag{100}$$

Rearranging:

$$\boxed{ae^{r_t} + ce^{-r_t} - fe^{R_t} - he^{-R_t} + \delta = \varepsilon_t,} \tag{101}$$

where:

$$\delta = b - g. \tag{102}$$

This is the basic econometric specification.

9.2 Parameter identification

The original six coefficients are:

$$\boldsymbol{\theta} = (a, b, c, f, g, h)'. \tag{103}$$

However, the equation contains b and g only through their difference.

Proposition 9.1. *The original structural equation cannot separately identify b and g.*

Proof. Consider:

$$b' = b + k, \qquad g' = g + k. \tag{104}$$

Then:

$$b' - g' = b - g. \tag{105}$$

The structural equation remains unchanged.

Therefore, the data cannot distinguish between the original pair and the transformed pair. □

The identifiable coefficient combinations are:

$$\boldsymbol{\theta} = (a, c, f, h, \delta)'. \tag{106}$$

There is also a common-scale indeterminacy in the exact homogeneous equation.
A normalization is therefore necessary.
For example:

$$a = 1. \tag{107}$$

This identifies coefficient ratios relative to a, provided the remaining design and statistical identification conditions hold.

9.3 Linear-in-parameters representation

Define:

$$\mathbf{X}_t = \begin{pmatrix} e^{r_t} \\ e^{-r_t} \\ -e^{R_t} \\ -e^{-R_t} \\ 1 \end{pmatrix}. \tag{108}$$

Then:

$$\mathbf{X}_t' \boldsymbol{\theta} = \varepsilon_t. \tag{109}$$

The model is nonlinear in the observed variables but linear in its structural coefficients.
This does not make ordinary least squares automatically applicable.
The equation is an implicit restriction without an independently specified dependent variable.
A normalization and a defensible statistical identification strategy are required.

9.4 Minimum-distance estimation

A natural estimator minimizes the structural residuals subject to normalization:

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \sum_{t=1}^{T} \left(\mathbf{X}_t' \boldsymbol{\theta}\right)^2. \tag{110}$$

Subject to:

$$a = 1. \tag{111}$$

Additional restrictions may include:

$$a, c, f, h > 0. \tag{112}$$

Care is required because minimizing residuals without a normalization can produce the trivial zero coefficient vector.

9.5 Generalized Method of Moments

Let $\mathbf{z}_t$ be a vector of instruments.

Assume:

$$E[\mathbf{z}_t \varepsilon_t] = \mathbf{0}. \tag{113}$$

Define the sample moment vector:

$$\mathbf{m}_T(\boldsymbol{\theta}) = \frac{1}{T} \sum_{t=1}^{T} \mathbf{z}_t \mathbf{X}_t' \boldsymbol{\theta}. \tag{114}$$

The GMM estimator is:

$$\boxed{\widehat{\boldsymbol{\theta}}_{\text{GMM}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\, \mathbf{m}_T(\boldsymbol{\theta})' \mathbf{W}_T \mathbf{m}_T(\boldsymbol{\theta}).} \tag{115}$$

Here, $\mathbf{W}_T$ is a positive-definite weighting matrix.

Instrument validity requires economic justification.

Lagged variables are not automatically valid instruments.

9.6 Maximum likelihood

Assume:

$$\varepsilon_t \sim N(0, \sigma_\varepsilon^2). \tag{116}$$

The log-likelihood is:

$$\begin{aligned} \ell = & -\frac{T}{2} \ln(2\pi) - \frac{T}{2} \ln(\sigma_\varepsilon^2) \\ & - \frac{1}{2\sigma_\varepsilon^2} \sum_{t=1}^{T} \left(\mathbf{X}_t' \boldsymbol{\theta} \right)^2 . \end{aligned} \tag{117}$$

The maximum-likelihood estimator maximizes this expression subject to normalization and parameter restrictions.

The Gaussian specification can be compared against Student-t and other heavy-tailed alternatives.

9.7 Conditional heteroskedasticity

Let:

$$\varepsilon_t = \sqrt{h_t} u_t. \tag{118}$$

Assume:

$$E[u_t] = 0, \qquad E[u_t^2] = 1. \tag{119}$$

A GARCH(1,1) specification is:

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}. \tag{120}$$

With:

$$\omega > 0, \tag{121}$$
$$\alpha \geq 0, \tag{122}$$
$$\beta \geq 0. \tag{123}$$

Covariance stationarity of the conventional GARCH(1,1) variance process requires:

$$\alpha + \beta < 1. \tag{124}$$

The complete model is:

$$\boxed{\mathcal{I}(r_t) = \mathcal{K}(R_t) + \sqrt{h_t} u_t.} \tag{125}$$

This permits investigation of volatility clustering in deviations from the structural relationship.

10 Simultaneity and Endogeneity

Return and risk may be jointly determined.

For example, changes in risk may affect required returns, while returns may affect measured volatility.

Let:

$$r_t = \alpha_0 + \alpha_1 R_t + \alpha_2 Z_t + u_t, \tag{126}$$
$$R_t = \beta_0 + \beta_1 r_t + \beta_2 W_t + v_t. \tag{127}$$

Here, Z_t and W_t are variables subject to exclusion restrictions.

The exclusion restrictions must be justified using economic theory.

The following distinctions are fundamental:

Structural relationship The mathematical equilibrium restriction.

Statistical relationship The conditional association in observed data.

Causal relationship The effect of an intervention under defensible identification assumptions.

The algebraic equation establishes the first. Econometric estimation investigates the second. Additional identification is required for the third.

11 Panel Econometrics and Structural Heterogeneity

Let i index assets, countries or institutions.

Let t index time.

The panel model is:

$$a_i e^{r_{it}} + c_i e^{-r_{it}} - f_i e^{R_{it}} - h_i e^{-R_{it}} + \delta_i = \varepsilon_{it}. \tag{128}$$

Three specifications are possible.

11.1 Homogeneous coefficients

$$\boldsymbol{\theta}_i = \boldsymbol{\theta}. \tag{129}$$

11.2 Heterogeneous coefficients

$$\boldsymbol{\theta}_i \neq \boldsymbol{\theta}_j. \tag{130}$$

11.3 Time-varying coefficients

$$\boldsymbol{\theta}_{it} = \boldsymbol{\theta}_{i,t-1} + \boldsymbol{\eta}_{it}. \tag{131}$$

The time-varying model permits the structural relationship to evolve across monetary regimes, financial crises and institutional changes.

Relevant methods include:

- Fixed-effects and random-effects estimators.
- Mean-group estimators.
- Common correlated effects.
- State-space models.
- Kalman filtering.
- Bayesian hierarchical estimation.
- Structural-break procedures.

12 Statistical Hypotheses

The theory generates several testable hypotheses.

12.1 Reflection symmetry

For return:

$$H_0 : a = c. \tag{132}$$

For risk:

$$H_0 : f = h. \tag{133}$$

12.2 Balance-point hypotheses

$$H_0 : r^* = 0. \tag{134}$$

$$H_0 : R^* = 0. \tag{135}$$

12.3 Structural stability

The null hypothesis of coefficient stability is:

$$H_0 : \boldsymbol{\theta}_t = \boldsymbol{\theta} \quad \forall t. \tag{136}$$

Alternative specifications may allow multiple structural breaks.

12.4 Branch dependence

Let:

$$S_t \in \{-, +\}. \tag{137}$$

The branch-selection hypothesis can be examined using latent-state models.

However, algebraic branches and statistically identified regimes must not be treated as identical without additional evidence.

13 Dynamic Theory

13.1 Continuous-time evolution

Let:

$$r = r(t), \qquad R = R(t). \tag{138}$$

The equilibrium condition is:

$$F(r(t), R(t)) = 0. \tag{139}$$

Differentiating with respect to time:

$$F_r \dot{r} + F_R \dot{R} = 0. \tag{140}$$

Therefore:

$$\boxed{\dot{R} = -\frac{F_r}{F_R} \dot{r}.} \tag{141}$$

This is the instantaneous dynamic compatibility condition along a regular equilibrium trajectory.

13.2 Adjustment dynamics

One possible adjustment mechanism is:

$$\dot{R} = -\lambda F(r, R), \qquad \lambda > 0. \tag{142}$$

Holding r fixed, consider an equilibrium R^* satisfying:

$$F(r, R^*) = 0. \tag{143}$$

Linearizing:

$$\dot{\widetilde{R}} = -\lambda F_R(r, R^*)\widetilde{R}. \tag{144}$$

Local asymptotic stability requires:

$$F_R(r, R^*) > 0. \tag{145}$$

Since:

$$F_R = -fe^R + he^{-R}, \tag{146}$$

the stability condition is:

$$\boxed{R^* < \frac{1}{2}\ln\left(\frac{h}{f}\right).} \tag{147}$$

Thus, under this particular adjustment rule, the lower risk branch is locally stable.
The upper branch is unstable.
This conclusion depends on the chosen adjustment mechanism.

13.3 Coupled adjustment dynamics

A more general system is:

$$\dot{r} = -\lambda_r F_r(r, R) + u_r(t), \tag{148}$$

$$\dot{R} = \lambda_R F_R(r, R) + u_R(t). \tag{149}$$

Here, $u_r(t)$ and $u_R(t)$ are external shocks or policy inputs.

The system can be studied using Jacobian eigenvalues, Lyapunov functions and numerical stability analysis.

The choice of adjustment equations is a separate economic hypothesis, not a consequence of the equilibrium equation alone.

14 Stochastic Extensions

Let return and risk evolve according to:

$$dr_t = \mu_r(r_t, R_t)\,dt + \sigma_r(r_t, R_t)\,dW_{r,t}, \tag{150}$$

$$dR_t = \mu_R(r_t, R_t)\,dt + \sigma_R(r_t, R_t)\,dW_{R,t}. \tag{151}$$

Here, $W_{r,t}$ and $W_{R,t}$ are Brownian motions.
Their instantaneous correlation is:

$$dW_{r,t}dW_{R,t} = \rho\,dt. \tag{152}$$

The stochastic dynamics permit investigation of return-risk co-movement.

If the exact equilibrium constraint is intended to hold at all times, the diffusion coefficients cannot be arbitrary.

The stochastic process must remain tangent to the equilibrium manifold, or the equilibrium condition must instead be interpreted as a long-run restriction or a stochastic residual condition.

14.1 Ito calculus

For a twice continuously differentiable function $G(r, R)$, Ito's formula gives:

$$dG = G_r\, dr + G_R\, dR \tag{153}$$

$$+ \frac{1}{2} G_{rr}(dr)^2 + \frac{1}{2} G_{RR}(dR)^2 + G_{rR}\, dr\, dR. \tag{154}$$

The quadratic variations are:

$$(dr)^2 = \sigma_r^2 dt, \tag{155}$$

$$(dR)^2 = \sigma_R^2 dt, \tag{156}$$

$$dr\, dR = \rho \sigma_r \sigma_R dt. \tag{157}$$

This permits stochastic analysis of the Irrational Return and Risk Functions.

14.2 Jump processes

Financial markets may experience discontinuous shocks.

A jump extension can be specified as:

$$dr_t = \mu_r dt + \sigma_r dW_{r,t} + \kappa_r dN_t, \tag{158}$$

$$dR_t = \mu_R dt + \sigma_R dW_{R,t} + \kappa_R dN_t. \tag{159}$$

Here, N_t is a counting process.

The jump components can represent abrupt market repricing, default events, geopolitical shocks or policy announcements.

15 Applications to Other Scientific Fields

15.1 Physics and mathematical physics

The exponential-reciprocal structure resembles hyperbolic energy functions.

The identity:

$$ae^r + ce^{-r} = 2\sqrt{ac}\cosh(r - r^*) \tag{160}$$

provides a mathematical connection with hyperbolic functions frequently encountered in mathematical physics.

The structural function has a unique minimum and grows in both directions.

This is analogous to certain potential-energy functions.

However, the financial function is not automatically a physical energy function.

Any physical interpretation requires a specified system, units and conservation laws.

15.2 Chemistry

Exponential functions appear in chemical kinetics, equilibrium constants and thermodynamic relationships.

For example, the Boltzmann factor contains an exponential dependence on energy and temperature.

The present theory may provide a mathematical analogy for studying reciprocal and exponential responses in chemical systems.

No direct chemical law follows from the financial equation without a separately derived physical model.

15.3 Biology

Biological growth, population dynamics and ecological responses frequently involve exponential functions.

The reciprocal component may be useful in modelling opposing growth and contraction mechanisms.

Potential applications include:

- Population growth and decline.
- Resource availability.
- Ecological resilience.
- Biological stress responses.
- Competing exponential processes.

Such applications would require domain-specific parameters and validation.

15.4 Psychology and behavioural economics

Investor decisions are influenced by:

- Loss aversion.
- Reference dependence.
- Risk perception.
- Ambiguity aversion.
- Overconfidence.
- Herd behaviour.

The reciprocal return component offers a possible mathematical representation of asymmetric responses to gains and losses.

Nevertheless, the model does not itself establish a psychological mechanism.

Behavioural hypotheses must be tested using experimental or observational evidence.

15.5 Psychiatry and decision-making

Risk perception, uncertainty tolerance and decision-making under uncertainty are also relevant to clinical and behavioural research.

The framework could be used as a mathematical representation of nonlinear responses to uncertainty.

It must not be used to infer psychiatric conditions or diagnose individuals.

15.6 Linguistics

The distinction between mathematical return and the linguistic term irrational is important.

A formal definition prevents semantic ambiguity.

The theory uses Irrational Return as a proper technical label rather than a claim about the arithmetic nature of the variable.

This is an example of the importance of operational definitions in scientific language.

15.7 Philosophy

The theory raises questions concerning:

1. Mathematical representation.
2. Realism and instrumentalism.
3. Equilibrium and disequilibrium.
4. Determinism and stochasticity.
5. Model identification.
6. Explanation and prediction.
7. Epistemic uncertainty.

A mathematical relationship does not establish that the corresponding economic mechanism exists in reality.

Empirical validation is necessary.

15.8 Religion and ethical philosophy

The framework can be examined alongside ethical questions concerning risk allocation, wealth distribution, responsibility and intergenerational welfare.

Religious and philosophical traditions may provide normative perspectives on these issues.

Such perspectives are distinct from positive mathematical propositions.

The theory does not derive theological conclusions from exponential functions.

16 Law, Institutions and Governance

16.1 Contractual risk allocation

Financial contracts distribute risk among parties.

The framework could be used to study how contractual parameters influence the distribution of return-risk exposure.

Relevant legal fields include:

- Contract law.
- Securities regulation.
- Banking regulation.
- Insolvency law.
- Insurance law.
- Fiduciary duties.
- Consumer financial protection.

Legal obligations cannot be inferred from the mathematical equilibrium condition.

The model may assist quantitative analysis, but legal conclusions require applicable statutes, regulations and case law.

16.2 Institutional economics

Institutions influence:

- Property rights.
- Contract enforcement.
- Financial transparency.
- Monetary credibility.
- Risk-sharing mechanisms.
- Capital allocation.

The coefficients may therefore be studied as institution-dependent parameters.

A panel specification can be constructed to investigate whether institutional characteristics explain cross-country heterogeneity.

17 Political Economy and International Relations

17.1 Sovereign financial risk

Let R_{it} represent a sovereign risk measure for country i at time t.

Let r_{it} represent a specified sovereign financial return measure.

The panel equation is:

$$a_i e^{r_{it}} + c_i e^{-r_{it}} - f_i e^{R_{it}} - h_i e^{-R_{it}} + \delta_i = \varepsilon_{it}. \tag{161}$$

Possible applications include:

- Sovereign bond markets.
- Foreign exchange reserves.
- Government financing costs.
- Sovereign credit spreads.
- External debt sustainability.
- Capital-flow volatility.

The model may be combined with conventional debt sustainability analysis.

It does not replace fiscal accounting identities or intertemporal government budget constraints.

17.2 Geopolitical risk

Geopolitical shocks can influence financial markets through:

- Trade disruption.
- Energy prices.
- Sanctions.
- Capital controls.
- Military expenditure.
- Currency instability.
- Sovereign borrowing costs.

A geopolitical risk variable G_t may be introduced into the structural coefficients:

$$a_t = a_0 + a_1 G_t. \tag{162}$$

Likewise:

$$f_t = f_0 + f_1 G_t. \tag{163}$$

Such specifications require sufficient variation and identification.

17.3 Diplomacy and international cooperation

International institutions can influence financial stability through information sharing, liquidity arrangements, monetary cooperation and crisis management.

The framework could be extended to a multi-country network.

Let:

$$\mathcal{G} = (V, E) \tag{164}$$

represent a network of countries or financial institutions.

The adjacency matrix is:

$$A_{ij} = \begin{cases} 1, & \text{if a specified relationship exists,} \\ 0, & \text{otherwise.} \end{cases} \tag{165}$$

Network exposure can be incorporated into the risk function.

For example:

$$R_i = R_i^{(0)} + \sum_{j=1}^{n} A_{ij}\omega_{ij}R_j. \tag{166}$$

This provides a potential framework for international financial contagion.

17.4 Welfare economics

A social welfare function can be introduced:

$$W = \sum_{i=1}^{n} \omega_i U_i. \tag{167}$$

Here, U_i represents individual or institutional welfare and ω_i represents a normative welfare weight.

The return-risk framework may contribute to measuring the financial consequences of alternative allocation decisions.

However, a welfare ranking requires explicit normative assumptions.

17.5 Warfare economics

Military expenditure, sovereign financing, trade disruption and emergency resource allocation can influence return and risk.

A development-defence allocation model might be written:

$$G_t + D_t = Y_t, \tag{168}$$

where:

G_t Development expenditure.

D_t Defence expenditure.

Y_t Available fiscal resources.

The theory can be used to investigate financial consequences of alternative fiscal allocations.

It does not determine military strategy or provide a substitute for humanitarian and international legal obligations.

18 Computer Science and Algorithms

18.1 Computational objective

The principal computational problem is to determine the admissible equilibrium pairs satisfying:

$$F(r, R) = 0. \tag{169}$$

The quadratic transformations provide closed-form solutions.

Numerical methods are useful when parameters vary over time or when the model is extended with additional nonlinear restrictions.

18.2 Quadratic solution algorithm

```
INPUT:
    a, b, c, f, g, h, R

CALCULATE:
    Phi = f*exp(R) + g + h*exp(-R)

    Delta = Phi^2 - 4*a*c

IF a == 0:
    Solve the resulting linear equation.
ELSE IF Delta < 0:
    Return no real solutions.
ELSE:
    x_plus = (Phi + sqrt(Delta))/(2*a)
    x_minus = (Phi - sqrt(Delta))/(2*a)

    Retain only positive roots.

    r_plus = log(x_plus)
    r_minus = log(x_minus)

OUTPUT:
    Admissible return branches.
```

A numerically stable implementation should avoid subtractive cancellation when computing the smaller quadratic root.

For positive a, c, a stable alternative can be constructed from the product of the roots.

18.3 Newton-Raphson method

For more general extensions, define:

$$F(r, R) = 0. \tag{170}$$

A Newton iteration in R, holding r fixed, is:

$$R_{k+1} = R_k - \frac{F(r, R_k)}{F_R(r, R_k)}. \tag{171}$$

The method requires:

$$F_R(r, R_k) \neq 0. \tag{172}$$

It may fail near turning points or singularities.
Bracketed methods should be used where possible.

18.4 Artificial intelligence and machine learning

Potential applications include:

- Nonlinear parameter estimation.
- Bayesian parameter inference.
- Regime classification.
- Branch classification.
- Financial anomaly detection.
- Forecasting residual distributions.
- Model comparison.

Machine learning may assist parameter estimation, but it should not replace identification analysis.

A neural network that predicts r from R does not automatically validate the structural theory.

A useful hybrid architecture would combine:

1. A structural exponential-reciprocal model.
2. A statistical residual model.
3. A regime-classification component.
4. An uncertainty-estimation component.

18.5 Bayesian extension

Let the parameter vector be:

$$\boldsymbol{\theta} = (a, c, f, h, \delta)'. \tag{173}$$

Bayesian inference gives:

$$p(\boldsymbol{\theta} \mid \mathbf{D}) \propto p(\mathbf{D} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}). \tag{174}$$

This permits uncertainty quantification and prior restrictions reflecting theoretically justified parameter domains.

19 Cybersecurity and Model Governance

Financial implementations require secure data acquisition and computational integrity.

Relevant considerations include:

- Data provenance.
- Authentication.
- Access control.
- Encryption.
- Audit logging.
- Reproducibility.
- Model version control.
- Supply-chain security.
- Adversarial data validation.

A compromised financial dataset may produce incorrect parameter estimates even if the mathematical implementation is flawless.

Model governance should therefore separate:

Data integrity Whether the observations are authentic and unaltered.

Model integrity Whether the implemented equations match the documented specification.

Computational integrity Whether the execution environment is reproducible.

Decision integrity Whether outputs are used within their validated scope.

The framework should be independently audited before deployment in regulated financial environments.

20 Interdisciplinary Research Matrix

Table 2: Potential interdisciplinary applications

Field	Principal connection	Research status
Algebra	Quadratic exponential transformation	Derived
Analysis	Convexity and derivatives	Derived
Geometry	Equilibrium manifolds	Derived
Economics	Return-risk compatibility	Proposed
Finance	Portfolio and volatility modelling	Proposed
Econometrics	Identification and estimation	Proposed
Statistics	Hypothesis testing	Proposed
Physics	Hyperbolic mathematical analogy	Analogical
Chemistry	Exponential equilibrium analogy	Analogical
Biology	Growth and contraction systems	Analogical
Psychology	Asymmetric risk perception	Hypothesis
Law	Contractual risk allocation	Application
Political economy	Sovereign risk	Application
International relations	Financial networks	Application
Welfare economics	Distributional consequences	Extension
Computer science	Computational solution methods	Derived
AI/ML	Hybrid structural estimation	Extension
Cybersecurity	Data and model integrity	Application

21 PGF/TikZ Vector Graphics

21.1 Structural architecture

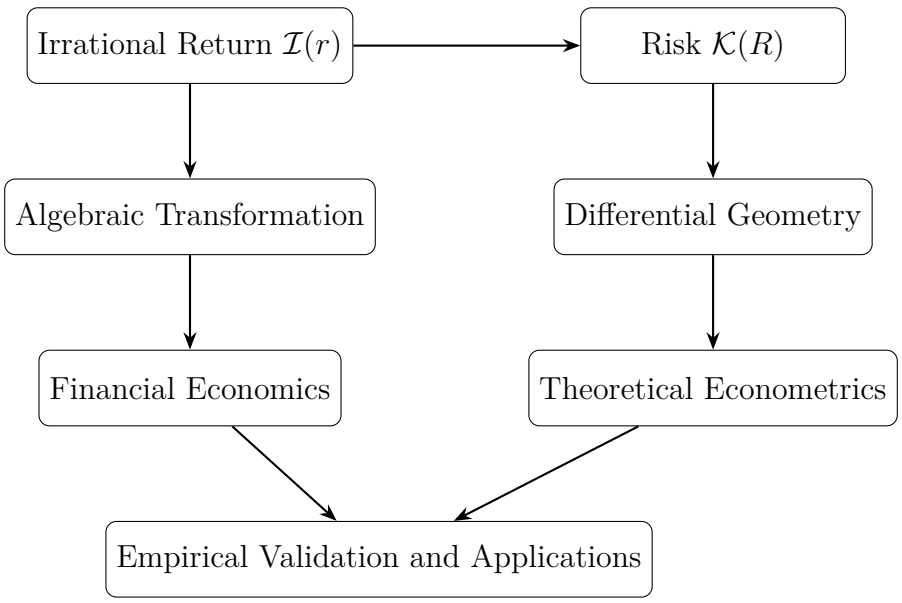


Figure 1: Conceptual architecture of the Theory of Irrational Return and Risk.

21.2 Equilibrium branches

The local geometry near a singular equilibrium can be illustrated by the two tangent directions.

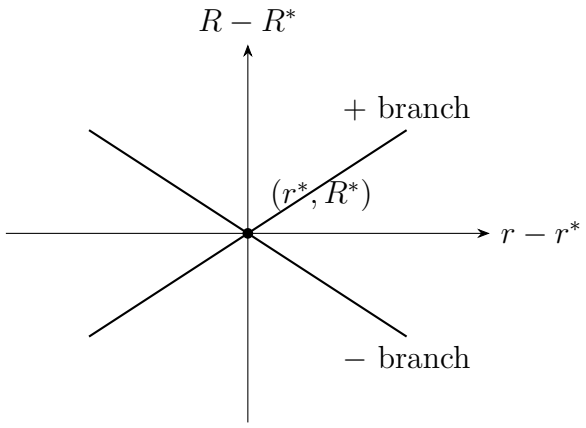


Figure 2: Local tangent geometry of the two equilibrium branches at a singular point.

22 Empirical Research Design

22.1 Data requirements

An empirical investigation requires observations on return and a clearly defined risk measure.

Potential datasets include:

- Equity index returns.
- Sovereign bond returns and yields.
- Exchange-rate returns.
- Cryptocurrency returns.
- Realized volatility.
- Conditional volatility.
- Credit spreads.
- Liquidity indicators.
- Macroeconomic variables.

Potential sources include central banks, statistical agencies, financial exchanges, international financial institutions and established market-data providers.

22.2 Research hypotheses

The following hypotheses should be evaluated:

1. The exponential-reciprocal specification provides an identifiable structural relationship.
2. The return and risk functions exhibit statistically significant nonlinearities.
3. The reflection-symmetry restrictions are empirically testable.
4. The structural coefficients may vary across financial regimes.
5. The theory may provide incremental explanatory or predictive information.

22.3 Benchmark models

The proposed model should be compared against:

- Linear return-risk regressions.
- Quadratic specifications.
- CAPM-based benchmarks.
- GARCH models.
- EGARCH and GJR-GARCH.
- Stochastic-volatility models.
- Regime-switching models.
- Flexible nonlinear statistical models.

22.4 Evaluation criteria

The comparison should include:

- In-sample fit.
- Out-of-sample forecast accuracy.
- Parameter stability.
- Residual diagnostics.
- Predictive log-likelihood.
- Information criteria.
- Calibration.
- Economic significance.
- Robustness to alternative risk measures.

An improvement in in-sample fit alone does not establish theoretical superiority.

23 Limitations and Epistemological Boundaries

Several limitations must be acknowledged.

First, the algebraic model does not establish a causal relationship between return and risk.

Second, the two branches do not automatically represent distinct investor populations.

Third, the structural balance points are mathematical minima rather than necessarily optimal financial choices.

Fourth, the exponential specification must be compared against competing functional forms.

Fifth, the parameters require normalization and sufficient statistical identification.

Sixth, a volatility proxy does not capture every dimension of financial risk.

Seventh, the proposed adjustment dynamics are additional hypotheses.

Eighth, interdisciplinary analogies do not establish physical, biological or psychological laws.

These distinctions are necessary to maintain scientific validity.

24 Further Theoretical Conjectures

Conjecture 24.1. *A sufficiently general time-varying extension of the Irrational Return–Risk framework may represent transitions between distinct financial equilibrium branches.*

Conjecture 24.2. *The introduction of institutional and network-dependent coefficients may permit the study of interconnected sovereign return-risk systems.*

Conjecture 24.3. *A hybrid structural-statistical architecture may improve the interpretation of nonlinear financial risk relative to an unrestricted predictive model, provided that empirical validation supports the additional structure.*

These conjectures require mathematical formalization and empirical investigation.

25 Conclusions

This treatise has established an exponential- reciprocal mathematical framework for the relationship between Irrational Return and Risk.

The fundamental equation:

$$\boxed{ae^{r} + b + ce^{-r} = fe^{R} + g + he^{-R}} \tag{175}$$

admits closed-form solutions in both directions.

The framework establishes:

1. Quadratic algebraic representations.
2. Conditions for real-valued solutions.
3. Two-branch solution structures.
4. Hyperbolic representations.

5. Structural balance points.
6. Strict convexity under positive coefficients.
7. Jacobian and Hessian properties.
8. Local invertibility conditions.
9. Singular equilibrium conditions.
10. Local branch-crossing geometry.
11. Comparative statics.
12. Financial volatility interpretations.
13. Econometric identification restrictions.
14. GMM and likelihood formulations.
15. Panel and dynamic extensions.
16. Stochastic and computational extensions.

The most important structural insight is that the relationship between return and risk need not be globally single-valued.

Depending on the structural coefficients, the equilibrium set may contain two disconnected branches or two branches intersecting at a singular balance point.

The theory therefore provides a framework for studying nonlinear financial relationships, equilibrium multiplicity, structural asymmetry and changing risk-return configurations.

Its ultimate contribution must be established through rigorous econometric identification, comparative empirical testing and out-of-sample validation.

A Structural Derivative Catalogue

The return function is:

$$\mathcal{I}(r) = ae^{r} + b + ce^{-r}. \tag{176}$$

Its derivatives are:

$$\mathcal{I}'(r) = ae^{r} - ce^{-r}, \tag{177}$$
$$\mathcal{I}''(r) = ae^{r} + ce^{-r}, \tag{178}$$
$$\mathcal{I}'''(r) = ae^{r} - ce^{-r}, \tag{179}$$
$$\mathcal{I}^{(4)}(r) = ae^{r} + ce^{-r}. \tag{180}$$

Therefore:

$$\mathcal{I}^{(n)}(r) = \begin{cases} ae^{r} + ce^{-r}, & n \text{ even}, \\ ae^{r} - ce^{-r}, & n \text{ odd}. \end{cases} \tag{181}$$

Likewise:

$$\mathcal{K}^{(n)}(R) = \begin{cases} fe^R + he^{-R}, & n \text{ even}, \\ fe^R - he^{-R}, & n \text{ odd}. \end{cases} \tag{182}$$

This provides a compact representation of all higher-order derivatives.

B Parameter Sensitivity

The structural balance points are:

$$r^* = \frac{1}{2}(\ln c - \ln a). \tag{183}$$

$$R^* = \frac{1}{2}(\ln h - \ln f). \tag{184}$$

Their parameter derivatives are:

$$\frac{\partial r^*}{\partial a} = -\frac{1}{2a}, \tag{185}$$

$$\frac{\partial r^*}{\partial c} = \frac{1}{2c}, \tag{186}$$

$$\frac{\partial R^*}{\partial f} = -\frac{1}{2f}, \tag{187}$$

$$\frac{\partial R^*}{\partial h} = \frac{1}{2h}. \tag{188}$$

The return minimum satisfies:

$$\mathcal{I}_{\min} = b + 2\sqrt{ac}. \tag{189}$$

Its parameter derivatives are:

$$\frac{\partial \mathcal{I}_{\min}}{\partial a} = \sqrt{\frac{c}{a}}, \tag{190}$$

$$\frac{\partial \mathcal{I}_{\min}}{\partial c} = \sqrt{\frac{a}{c}}, \tag{191}$$

$$\frac{\partial \mathcal{I}_{\min}}{\partial b} = 1. \tag{192}$$

Similarly:

$$\frac{\partial \mathcal{K}_{\min}}{\partial f} = \sqrt{\frac{h}{f}}, \tag{193}$$

$$\frac{\partial \mathcal{K}_{\min}}{\partial h} = \sqrt{\frac{f}{h}}, \tag{194}$$

$$\frac{\partial \mathcal{K}_{\min}}{\partial g} = 1. \tag{195}$$

These expressions quantify how structural parameter changes alter the balance points and minimum values.

C Numerical Validation Protocol

A computational implementation should verify:

1. Algebraic roots.
2. Positivity of exponential roots.
3. Reconstruction of original equations.
4. Discriminant conditions.
5. Derivative calculations.
6. Parameter normalization.
7. Numerical stability.
8. Branch continuity.
9. Boundary behaviour.
10. Reproducibility.

For every admissible numerical solution, calculate:

$$\mathcal{R} = \left| \mathcal{I}(r) - \mathcal{K}(R) \right|. \tag{196}$$

A numerical solution is accepted only when:

$$\mathcal{R} < \epsilon, \tag{197}$$

where ϵ is a specified numerical tolerance.

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Glossary

Branch One of the possible mathematical solutions of the equilibrium equation.

Convexity A property whereby the second derivative of a univariate function is non-negative.

Discriminant The expression determining the number and nature of roots of a quadratic equation.

Equilibrium Manifold The set of pairs (r, R) satisfying the structural equality.

Exponential-Reciprocal Function A function combining positive and negative exponential components.

Generalized Method of Moments An econometric estimation method based on population moment restrictions.

Hessian The matrix of second-order partial derivatives.

Irrational Return The named return variable of the proposed theory.

Jacobian The matrix of first-order partial derivatives.

Local Invertibility The property that permits one variable to be expressed locally as a function of another.

Risk The risk variable represented by R, whose operational meaning must be specified in each application.

Singular Equilibrium An equilibrium point at which both first partial derivatives of the structural function vanish.

Structural Balance Point The unique minimizer of the corresponding strictly convex exponential-reciprocal function.

Structural Identification The ability to recover model parameters or parameter combinations from observable information and restrictions.

Volatility A statistical measure of variability in financial returns.

The End